Assumptions Validation - Probe System

Summary: Key Findings: ✅ Scientific method tool exists and works ✅ Being system has security measures to replicate ✅ Reality system exists and supports Beings ❌ Learning algorithm not...

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What Happened

> Key Findings: ✅ Scientific method tool exists and works ✅ Being system has security measures to replicate ✅ Reality system exists and supports Beings ❌ Learning algorithm not defined (CRITICAL) ⚠️ Observation API unclear (MEDIUM) ❓ D&D integration unclear (MEDIUM).

Recommendation: MUST FIX - Define learning algorithm before implementation.

> Date: 2026-01-14 09:55:39 PST Context: OriginPoint Probe Experimental System Plan Validation Method: Multi-Source Evidence-Based.

Evidence: ✅ No permission setting: `scientific_method_tool/state_capture.py:68` creates directory without setting permissions ✅ No file permissions: `_save_state()` method (not shown but implied) doesn't set permissions ✅ Comparison: Being system sets permissions, StateCapture doesn't.

Recommendation: Design phase transition logic Implement pattern recognition Test each phase separately.

Evidence: ✅ Directory creation: Python `Path.mkdir(parents=True, exist_ok=True)` works ✅ Being system pattern: Being system creates `_hidden/.truth/beings/` successfully ❓ Permissions: Not clear if directory will have correct permissions ❓ Disk space: Not checked.

> Recommendation: Add directory creation with permission setting Check disk space before writes Follow Being system pattern.

Evidence: ✅ Being spawning: `BeingSystem.spawn_being()` exists ✅ Reality integration: Beings are added to Reality's `beings` list ✅ Spawning pattern: Pattern exists for spawning entities into Realities ⚠️ Probe-specific: Not clear if Probe needs special spawning logic.

> Implement Exploration Phases: Design and test phase transitions Random phase Pattern recognition Systematic phase Adaptive phase.

Design Feedback Loops: Define pressure detection and response generation Design Reflection System: Define analysis algorithm and insight generation Test Spawning Process: Verify Probe can spawn into Reality.

Validation Method: File system check, code analysis.

Recommendation: Design analysis algorithm Implement cause-effect detection Test insight generation.

> Evidence: ✅ File permissions: `src/waft/being.py:1964` sets `0o600` for files ✅ Directory permissions: `src/waft/being.py:1444` sets `0o700` for directories ✅ Path validation: `src/waft/being.py:1949` has `_validate_being_id()` method ✅ Path traversal protection: `src/waft/being.py:1955` validates paths are within project ✅ Security comments: Code has "CRITICAL" security comments.

Critical Assumptions: 3 ✅ 1 proven ⚠️ 1 partially proven ❓ 1 insufficient evidence.

> Category: Dependency Risk: Critical Status: ✅ PROVEN Confidence: 1.0.

Category: Code Risk: Critical Status: ✅ PROVEN Confidence: 1.0.

Category: System Risk: Critical Status: ✅ PROVEN Confidence: 1.0.

Assumption Statement: The `scientific_method_tool/` exists and can be used for hypothesis formation and testing.

> Evidence: ✅ Tool exists: `scientific_method_tool/` directory found ✅ Components exist: `hypothesis.py`, `state_capture.py`, `data_collection.py`, `experiment.py`, `experiment_loop.py`, `analysis.py` ✅ Implementation complete: `IMPLEMENTATION_SUMMARY.md` shows all components implemented ✅ Exports available: `__init__.py` exports all components ✅ Usage examples: `example_usage.py` and `README.md` show usage.

Assumption Statement: The Being system has security measures (file permissions, path validation) that should be replicated for Probe system.

> Evidence: ✅ Reality system exists: `src/waft/reality.py` found ✅ Reality class: `Reality` class defined with `beings` list ✅ RealitySystem: `RealitySystem` class manages Realities ✅ Being integration: Reality has `beings: List[str]` attribute ✅ Reality types: Multiple RealityType enums (LEARNING, TESTING, EVOLUTION, etc.).

Assumption Statement: Scientific method tool's StateCapture doesn't set file permissions (security gap).

Evidence: ✅ Reality class exists: `Reality` class has state (configuration, beings list) ✅ Being class exists: `Being` class has observable state (skills, fitness, state) ❓ Observation API: No explicit observation methods found in Reality or Being classes ❓ Environmental features: Not clear what "environmental features" means in Reality context.

Recommendation: Check if Reality/Being expose observable state Define what "environmental features" means Test observation capabilities before implementation.

> Evidence: ✅ Personality system exists: `Being` class has `personality` dict and `personality_type` ❓ D&D integration: No existing D&D character system found ❓ Enhancement mechanism: Not clear how stats would enhance personality ❓ Conflict resolution: No evidence of how conflicts would be resolved.

Recommendation: Design D&D stat-to-personality mapping Define enhancement rules (additive vs. override) Test with sample personalities.

> Recommendation: Design communication protocol Define suggestion/review format Test with sample interactions.

Evidence: ❓ Phase logic: Not implemented, logic undefined ❓ Pattern recognition: Not implemented, algorithm undefined ❓ Adaptive combination: Not implemented, mechanism undefined.

Evidence: ❓ Pressure detection: Not implemented ❓ Response generation: Not implemented ❓ Feedback analysis: Not implemented.

Recommendation: Design pressure detection algorithm Implement response generation Test feedback analysis.

> Assumption Statement: Probe can update behavior based on experiment results (learning algorithm exists).

Evidence: ❌ No learning algorithm: Plan doesn't specify learning algorithm ❌ No update mechanism: Plan doesn't specify how behavior is updated ❌ No convergence criteria: Plan doesn't specify when learning is complete.

> Recommendation: Define learning algorithm before implementation Specify update mechanism Define convergence criteria.

Evidence: ✅ Reality class exists: Has observable state ✅ Being class exists: Has observable state ❓ Observation API: No explicit observation methods ❓ Environment features: Not defined.

Evidence: ❓ Analysis algorithm: Not defined ❓ Cause-effect detection: Not implemented ❓ Insight generation: Not implemented.

Recommendation: Use existing spawning pattern Check if Probe needs special logic Test spawning process.

> Evidence: Plan doesn't specify learning algorithm No update mechanism defined No convergence criteria.

Evidence: Reality and Being classes exist No explicit observation methods found Environmental features not defined.

> Define Learning Algorithm: Specify how Probe learns from experiments Update mechanism Convergence criteria Learning rate/parameters.

Design D&D Integration: Define how D&D stats enhance personality Stat-to-personality mapping Enhancement rules (additive vs. override) Conflict resolution.

Additional Details

Test Scientific Method Integration: Verify tool works with Probe data Test with sample Probe data Check for limitations Verify integration.

Design Collaborative Piloting: Define Probe-AI communication protocol Suggestion format Review process Error handling.

`src/waft/being.py:1964` - File permissions (0o600) `src/waft/being.py:1949` - ID validation `src/waft/reality.py:34` - Reality class `scientific_method_tool/state_capture.py:68` - State capture (no permissions) `scientific_method_tool/__init__.py` - Tool exports.

> `scientific_method_tool/README.md` - Tool documentation `scientific_method_tool/IMPLEMENTATION_SUMMARY.md` - Implementation status.

Overall Assessment: Most infrastructure exists, but core Probe functionality (learning, observation, D&D integration) needs definition before implementation.

> Recommendation: Define learning algorithm, validate observation API, and design D&D integration before proceeding with implementation.

This validation uses evidence from code analysis, file system checks, and plan review. All conclusions are traceable to specific evidence sources.

Conclusion: ✅ PROVEN - Scientific method tool exists and is fully implemented.

Conclusion: ✅ PROVEN - Being system has proper security measures that should be replicated.

> Recommendation: Replicate Being system security measures in Probe system.

Assumption Statement: Reality system exists and can host Beings (and by extension, Probes).

> Conclusion: ✅ PROVEN - Reality system exists and supports Beings.

Recommendation: Probe can use existing Reality system. Verify Reality observation API.

Category: System Risk: Medium Status: ⚠️ PARTIALLY PROVEN Confidence: 0.6.

Conclusion: ⚠️ PARTIALLY PROVEN - Reality and Being classes exist, but observation API is unclear.

> Assumption Statement: D&D stats can enhance existing personality system without conflicts.

Conclusion: ❓ INSUFFICIENT EVIDENCE - Personality system exists, but D&D integration is unclear.

> Validation Method: Code analysis (no evidence found).

Assumption Statement: Exploration can evolve from random → systematic → adaptive correctly.

Validation Method: Code analysis (no evidence found).

Conclusion: ❓ INSUFFICIENT EVIDENCE - Exploration phases not implemented.

> Assumption Statement: External/Internal pressure loops work as designed.

Validation Method: Code analysis (no evidence found).

> Conclusion: ❓ INSUFFICIENT EVIDENCE - Feedback loops not implemented.

Category: System Risk: Medium Status: ⚠️ PARTIALLY PROVEN Confidence: 0.7.

Assumption Statement: `_experiments/probe/` directory can be created and is writable.

Validation Method: Code analysis, pattern matching.

> Conclusion: ⚠️ PARTIALLY PROVEN - Directory creation works, but permissions unclear.

Assumption Statement: Scientific method tool can handle Probe-specific data structures.

> Category: System Risk: Medium Status: ⚠️ PARTIALLY PROVEN Confidence: 0.5.

Assumption Statement: Observation system can observe Reality, Beings, environment.

Conclusion: ⚠️ PARTIALLY PROVEN - Classes exist, but observation API unclear.

Assumption Statement: Reflection system can analyze feedback loops and cause-effect relationships.

> Validation Method: Code analysis (no evidence found).

Conclusion: ❓ INSUFFICIENT EVIDENCE - Reflection system not implemented.

> Category: System Risk: Medium Status: ✅ PROVEN Confidence: 0.9.

Conclusion: ✅ PROVEN - Spawning pattern exists, Probe can use it.

> Recommendation: Validate observation API before implementation.

Evidence: Personality system exists No D&D system found Enhancement mechanism undefined.

> Recommendation: Design D&D integration before implementation.